Purabh Singh

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Education

Rutgers University Aug 2023 – May 2027

B.S. in Computer Science, Minor in Data Science *New Brunswick, NJ*

- **Relevant Coursework:** Data Structures, Algorithms, Computer Architecture, Discrete Structures I & II, Intro to AI, Systems Engineering, Data Management for Data Science
- **Achievements:** MusaLabs Hackathon Winner (1st of 50+ teams) for LeaseLink, a full-stack housing-matching app · Dean's List

Experience

Software Engineering Intern June 2026 – Aug 2026

BNY *Pittsburgh, PA*

- Built an immutable fintech payments audit trail on a private Hyperledger Besu blockchain (Solidity, Web3j), validating and recording $2B+ in cleared debits and credits for real-time reconciliation across 3 fragmented systems with zero payment-latency impact, tested end-to-end using Apache ActiveMQ Artemis to simulate live message-queue traffic, fronted by an Angular/ethers.js UI that reads directly from Besu for live ledger visibility during backend downtime.
- Developed a real-time SWIFT MT/MX "blotter" dashboard for an Agile payments team, tracing every message as it moved through the platform's Java/Spring Boot systems, delivering first-ever end-to-end visibility into message flow across 15M+ daily messages.
- Built an Angular observability and troubleshooting dashboard for cross-functional ops teams, pulling from Splunk logs to monitor the health and security posture of all 11 payment-platform systems, surfacing per-system vulnerability counts and month-over-month trends and auto-generating Jira tickets from flagged issues to cut manual triage time by 60%.

Software Engineering Intern May 2025 – Aug 2025

Essex Securities *Jersey City, NJ*

- Built an AI agent using LangChain and RAG (pgvector semantic search, FastAPI endpoints) to automate extraction and comparison of actual vs estimated EPS from quarterly earnings reports, cutting manual review time by over 70%.
- Built an MVP BTC-USD spot exchange in Python (FastAPI, asyncio, PostgreSQL) with a price-time-priority matching engine supporting limit/market orders, partial fills, self-trade prevention, and sequenced WebSocket market data.
- Designed an append-only double-entry ledger with atomic per-trade settlement, pre-trade risk holds, and crash recovery that rebuilds the order book, with automated invariant checks proving every balance reconciles.
- Automated limit order execution via Bloomberg EMSX API integration, reducing manual entry time by 20%.

Software Engineering Intern May 2024 – Aug 2024

Azonix Tech Private Ltd *Remote*

- Built a real-time facial-recognition pipeline processing a live 30 FPS video feed at the edge, using OpenCV and an ONNX-optimized InsightFace embedding model to detect and identify enrolled individuals in under 45 ms per frame at 96%+ accuracy.
- Engineered an IoT telemetry platform ingesting 5,000+ MQTT messages/sec from 500 concurrent sensor devices, persisting time-series data to TimescaleDB for sub-second querying and streaming anomaly detection.
- Trained an ML (XGBoost) predictive-maintenance model on 100K+ rows of device telemetry to forecast equipment failures roughly 6 hours in advance, reaching 0.92 AUC and cutting projected unplanned downtime by 40%.

Projects

FastRust | *Rust, Tokio, Hyper, OpenAPI, Criterion* Feb 2026 – Jun 2026

- Built a FastAPI-style Rust web framework from scratch: a trie router, typed extractors (Path, Query, JSON, State), procedural route macros, and onion-style middleware (CORS, bearer auth, logging).
- Generated OpenAPI 3.1 docs and a Swagger UI straight from route metadata; benchmarked with Criterion at 172 ns route lookups and under 2.3 μs per request.

Mochee | *TypeScript, Node.js, PostgreSQL, WebSockets, React Three Fiber* Oct 2026 – Present

- Built a 3D AI SAT/PSAT tutor with two modes: **Study**, where students upload their own notes and get answers citing the exact page and quote (abstaining when the notes do not cover it), and **Ranked**, Elo-rated 1v1 duels on a self-verifying question bank.
- Built a real-time duel engine over WebSockets where the server owns the clock and scoring, with Elo updates in row-locked Postgres transactions and exactly-once payouts via idempotency keys, publishing each generated question only after an independent re-solve, code-checked math and near-duplicate rejection; 100+ backend tests.

vaultdb | *Python, Ollama, SQLite, MCP* Apr 2026 – Present

- Built a local-first knowledge base and MCP server that parses an Obsidian vault into code-aware chunks, embeds them with Ollama, and serves hybrid keyword + vector search from a SQLite index to Claude Code through 4 MCP tools.
- Kept the index live with a file watcher, capped code-aware chunks to fit the embedding model's context window, and covered the parser, search, watcher and MCP tools with 98 tests.

Technologies

Languages: Python, Java, TypeScript, JavaScript, Rust, C++, C, SQL | **Backend**: Node.js, Spring Boot, FastAPI, Express, NestJS, Kafka, ActiveMQ Artemis, REST, WebSockets, MQTT, OAuth | **Data**: PostgreSQL, MySQL, SQLite, MongoDB, TimescaleDB, BigQuery, Qdrant, pgvector | **Frontend**: React, Next.js, Angular, Tailwind CSS, Tauri, Swift | **AI/ML**: LangChain, Ollama, scikit-learn, XGBoost, OpenCV, Pandas | **Cloud/DevOps**: AWS, Azure, Docker, Kubernetes, CI/CD, Splunk | **Blockchain**: Solidity, Hyperledger Besu, Web3j, ethers.js